Shubhajit Roy

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Education

Indian Institute of Technology Gandhinagar **July 2022 – Present**
Ph.D in Computer Science and Engineering Gandhinagar, India
CPI - 9.14

Indian Association for the Cultivation of Science **October 2020 – July 2022**
M.Sc in Mathematics and Computing with Major in Computer Science Kolkata, India
CGPA - 8.42

Presidency University **June 2017 – August 2020**
B.Sc in Mathematics Kolkata, India
CGPA - 6.95

Experience

Research Intern - Platforms in AI **May 2026 – August 2026**
IBM Research Bangalore, India

Publications

- "T-SNN: Temporal Simplicial Neural Network for EEG Decoding" ArXiv (2026). Nikita Malik*, Shubhajit Roy*, Mohit Kataria*, Isuru Herath, Suraj Yadav, Inés García-Redondo, and Dhananjay Bhaskar.
- "SiST-GNN: Simultaneous Spatial-Temporal Message Passing for Dynamic Graph Representation Learning" ArXiv (2026). Shubhajit Roy and Anirban Dasgupta.
- "FIT-GNN: Faster Inference Time for GNNs that 'FIT' in Memory Using Coarsening" TMLR (2026), AutoML 2026 (Hot-off-the-press track), LoG 2026 (TMLR track). Shubhajit Roy*, Hrriday Ruparel*, Kishan Ved and Anirban Dasgupta.
- "Local Fragments, Global Gains: Subgraph Counting using Graph Neural Networks" DiffCoAlg @ NeurIPS (2025), WEB&GRAPH @ WSDM 2026. Shubhajit Roy, Shrutimoy Das, Binita Maity, Anant Kumar, and Anirban Dasgupta.

Software

simplicial-sindy (PyPI) **Dr. Dhananjay Bhaskar**
July 2026 - v0.1.0, MIT License

- Python package that discovers simplicial complexes (nodes, edges, triangles) from multivariate time series by fitting a sparse polynomial dynamical model (SINDy) with an ADMM group-lasso under a second-order-cone simplicial hierarchy constraint, exposing higher-order (triadic) interactions rather than pairwise ones only.

Projects

Graph Representation Learning for Fast Detector Simulation **Prof. Shanmuganathan Raman**
January 2024 - April 2024

- Quark and gluon jets from high-energy particle collisions produce distinct patterns in jet images, but distinguishing them remains a core challenge in particle physics.
- This project applies machine learning, especially GNNs, VAEs, and GVAEs, to detect patterns that differentiate quark- from gluon-origin jets.

Graph Neural Network for weather data prediction **Prof. Anirban Dasgupta**
February 2023 - Present

- Given a set of locations as nodes and edges between, our task is to predict different weather stats based on past information.
- We propose to use GNNs in this streaming fashion and still overcome the current problems faced while learning from streaming graphs.
- One major problem tackled is the catastrophic forgetting problem, where the GNN model tends to forget past patterns in the graphs. Another problem faced is the memory consumption of these models.

Joint Entity Recognition and Relation extraction **Dr. Partha Basuchowdhuri**
September 2021 - Present

* denotes equal contribution (co-first authors).

- Given a sentence, we need to find the entities in the sentences and the relation between these entities. This task helps us in the construction of a Knowledge Graph where the nodes are the entities, and the edges between these nodes have the corresponding relation.
- In our proposed model, we intend to use GNNs to extract the dependencies between the words in a sentence and then predict the entities and relations.

MedEx - A Software for an Pharmacy database **Prof. Paramartha Dutta**

June 2021 - July 2021

- Implemented the SRS and the prototype code for transaction management of an online pharmaceutical database.
- The software helps the customer of the wholesaler to open an account and do all the transactions in a digital method. It includes facilities like credit payment, seeing transaction history, etc.
- Done along with Sarthak Das and Avishek Lahiri as a part of the IACS Computer Science Unit 2020 batch.

IACS Neural Networks Verification System (INNVerS) **Dr. Rajarshi Ray**

March 2021 - August 2021

- A tool for verification of ReLU neural network (for example, Acas-Xu) properties.
- The tool uses SCIP for optimizing and finding the bound with respect to the neural network and z3 for checking satisfiable conditions.
- Done along with Sarthak Das and Avishek Lahiri as a part of the IACS Computer Science Unit 2020 batch.

Roles and Responsibilities

Tutorial - ACM Summer School: AI for Social Good (Graph Neural Networks: Hands On) **June 2025**
Indian Institute of Technology Gandhinagar, India Slides

Reviewer - Learning on Graphs (LoG) Conference **August 2026**

Reviewer - ACM Transactions on Knowledge Discovery from Data **June 2026**

Reviewer - Empowering Global South AI @ AAAI 2026 **December 2025**

Reviewer - GCLR @ AAAI 2026 **October 2025**

Reviewer - NeurIPS 2025 **July 2025**

Ethics Reviewer - NeurIPS 2025 **July 2025**

Ethics Reviewer - NeurIPS 2024 **August 2024**

Ethics Reviewer - NeurIPS 2023 **July 2023**

Volunteer - Foundations of Software Technology and Theoretical Computer Science **December 2024**
Indian Institute of Technology Gandhinagar, India

Volunteer - Randomized Methods For Approximation And Parameterized Algorithms **December 2022**
Indian Institute of Technology Gandhinagar, India

Volunteer - Indian Symposium on Machine Learning **December 2022**
Indian Institute of Technology Gandhinagar, India

Teaching Assistant - Scalable Data Science **July 2025 - November 2025**
NPTEL

Teaching Assistant - Introduction to Data Science **January 2025 - May 2025**
Indian Institute of Technology Gandhinagar, India

Teaching Assistant - Introduction to Data Science **January 2023 - May 2023**
Indian Institute of Technology Gandhinagar, India

Teaching Assistant - Ordinary and partial differential equations **August 2022 - November 2022**
Indian Institute of Technology Gandhinagar, India

Relevant Coursework

- Social and Complex Networks
- Machine Learning
- Optimisation in ML
- GNN and Transformers
- Understanding Large Language Models

Events Attended

ACM SIGMOD/PODS **May 2026**
Bangalore, India

Indian Symposium on Machine Learning **December 2025**
BITS Pilani Hyderabad Campus, India

Frontiers of Graph Algorithms **December 2025**
Indian Institute of Science, Bangalore, India

Neuroscience, Data Science and Dynamics **April 2025**
International Center of Theoretical Sciences, Bangalore, India

Indian Symposium on Machine Learning **December 2024**
BITS Pilani Goa Campus, India

Pingala Interactions in Computing 2024 **February 2024**
Infosys Ltd Mysore, India

CODS-COMAD 2024 **January 2024**
International Institute of Information Technology Bangalore, India

Indian Symposium on Machine Learning **December 2023**
Indian Institute of Technology Bombay, India

Data Science: Probabilistic and Optimization Methods **July 2023**
International Center of Theoretical Sciences, Bangalore, India

Technical Skills

Languages: Python, Java, C, C++, R, SQL, Shell, LaTeX
Technologies/Frameworks: PyTorch, TensorFlow, GitHub
Communicating Language: English, Hindi, Bengali

Certificate and online Courses

Certification in Scientific Writing **April 2025**
Indian Institute of Technology Gandhinagar

- The Certificate in Scientific Writing is a professional credential that validates a participant's proficiency in writing clear, accurate, and well-structured scientific documents. It demonstrates the holder's ability to communicate research effectively in academic and professional settings.

Introduction to IoT **August 2020**
Cisco Networking Academy

- Learned how the current digital transformation is creating unprecedented economic opportunity. Understood how the IoT is bridging the gap between operational and information technology systems. Discovered how standard business processes are being transformed.

Elements of AI **June 2020**
University of Helsinki

- It provides a gentle introduction to the basics that enables us to understand what is possible, and what isn't, and to recognise how AI is affecting our lives.
- We can define and discuss AI, be critical about AI-related claims, explain the underlying technologies including machine learning and neural networks and understand the major implications of AI.

Awards and Achievements

Best Poster Award - PhD Research Showcase **January 2025**
Indian Institute of Technology Gandhinagar

UGC-NET 2022 **November 2022**
NTA UCG

- Qualified for JRF & Assistant Professor

GATE 2022 **March 2022**

Hacktoberfest 2022 **October 2022**
Github and Digital Ocean

- Successfully completed Hacktoberfest, 2022, by contributing in public Github repository.

Hacktoberfest 2021 **October 2021**
Github and Digital Ocean

- Successfully completed Hacktoberfest, 2021, by contributing in public Github repository.
- Four pull requests were accepted by a public Github repository - **Cosmos**

Digital Wellness Online Challenge 2015 **July 2015**

National e-governance Division